

Electronics 101 — from the dev board to the class board

Lecture 2 • Fri 18 Sep 2026 • 14:20–16:20 • R311 IAMS

Six ideas on our schematic • KiCad's six operations • place your missing parts • route your zone • your first PR into the class board

Where we are

- Every phone talks to its ESP32 (E2 merged). Every student has a block and a `SPEC.md` with a number.
- The **full schematic** exists (PDF on the site). Your **gapped sheet** is missing 3–4 parts. The **PCB** has every footprint pre-placed and your rule area `ZONE_B<N>` with your name on it.
- **JLC assembles everything.** You solder nothing. The dev board stays the fallback.
- **Order: Mon 28 Sep.** PR Wed 23 • review Fri 25 • fixes Sun 27. Slip here = slip everywhere.

The board, v0.6 — one block per student

Block	What is in your zone	Parts	Route
B1	ADS8688 + $8 \times (1 \text{ k}\Omega, 1 \text{ nF}, \text{BAV99}) \rightarrow \text{AI1-8}, \pm 10 \text{ V}$ 16-bit	≈ 32	1.5 h
B2	USB-C, jack, ORing, polyfuses, TVS, $2 \times 2 \text{ W}$ isolated $\rightarrow \pm 12 \text{ V}$, 78L05, AMS1117	≈ 20	1 h
B3	DAC8563 + OPA2192 $\pm 10 \text{ V}$ + $2 \times (49.9 \Omega + \text{clamp}) \rightarrow \text{AO1}, \text{AO2}$	≈ 15	1 h
B4	4 relays via ULN2003, $2 \times 6\text{N}137$, terminals, LEDs	≈ 24	1.25 h
B5	74AHCT541 ($8 \times \text{TTL}$), 74HCT125, 74LVC1T45 (TRIG), TCXO (DNP)	≈ 15	1 h

Base (instructor): socket, buses, LED, Qwiic, header, star point, front panel 3×4 SMA at 20 mm.

(a) Rails, decoupling, ground — B2

- 5 V in (USB-C **or** jack, SS34 ORing) → **+5V_RAW** → AMS1117 → **+3V3**
- $2 \times$ B0512S-2WR3 isolated modules → **+12 V / AGND / -12 V** → 78L05 → **+5VA**
- **Rule:** anything not voltage-sensitive runs from raw 5 V. ± 12 V and +5VA feed only op-amps, ADC, DAC.
- **100 nF at every supply pin, 10 μ F per rail per zone.** Distance beats value.
- **Two grounds, one star point.** GND (power, logic; layer 2 plane) and AGND (signals) meet once, at the B1/base boundary.

(b) What $\pm 10\text{ V}$ and 16 bits buy — B1 / B3

- $20\text{ V} / 65,536 = 0.3\text{ mV per LSB}$
- Every other choice on the board exists to keep noise below that: the star point, AGND copper under the ADC, no digital under the inputs, SPI entering from one side.
- Input network **1 k Ω $\rightarrow$ 1 nF $\rightarrow$ BAV99 to $\pm 12\text{ V}$** : current limit, gentle low-pass ($\approx 160\text{ kHz}$), clamp. A 30 V accident flows into the rail, not the chip.
- **Sampling**: 500 kSPS aggregate; anything above half the sample rate aliases. The Scope tab: rolling tens of kS/s (WiFi-limited), burst up to the ADC.

(c) The op-amp stage as a black box — B3

DAC8563: 0–5 V, 16-bit, 2.5 V reference out
OPA2192 as a **difference amplifier**, 10 k Ω /
40 k Ω :

$$AO = 4(V_{DAC} - 2.5 \text{ V}) \Rightarrow -10 \dots + 10 \text{ V}$$

- rails wider than the swing ($\pm 12 \text{ V}$)
- **49.9 Ω** before the SMA: the cable is a capacitor; the scope's 50 Ω input halves what you see
- BAV99 clamp after the resistor
- "clipped" = the output hit its rail — B3's alarm

(d) Isolation — B4

- A **relay**: coil and contacts share no copper. ULN2003 drives four coils from GPIO; its COM pin carries the flyback diodes that eat the coil's kick.
- An **optocoupler** (6N137): LED and transistor share no copper; 5–24 V in, 3.3 V out, fast.
- **Creepage**: the distance that keeps it so — **2.5 mm band**, no copper of any other net under it, on any layer. DRC checks it; your reviewer checks it.
- Contacts: $\leq 30\text{ V DC}$, **1 A. Never mains.**

(e) 3.3 V vs 5 V logic — B5

- ESP32-S3 outputs 0 / 3.3 V and must **never** see 5 V on a pin.
- Lab equipment speaks **5 V TTL**: high ≥ 2 V, outputs up to 5 V.
- **74AHCT541**: 3.3 V in $\rightarrow$ 5 V out, eight lines (6 DIO + 2 fast).
- **74LVC1T45**: translates either way under a direction pin $\rightarrow$ the **TRIG** SMA is input *or* output. 100 Ω in series: two drivers on one wire survive.
- TTL into 50 $\Omega \approx 2.5$ V; into 1 M $\Omega \approx 5$ V — the note on the panel.

(f) The buses that tie it together — base

SPI — three shared wires (SCLK 12, MOSI 11, MISO 13) + one chip-select per device: ADC `/cs` 10, DAC `/SYNC` 5. 33 Ω series on the data lines; short, one direction from the socket.

I²C — two wires (SDA 1, SCL 2), 4.7 k Ω pull-ups, addresses: OLED 0x3C, two Qwiic connectors for anything else.

Every block's driver is "write a few bytes on one of these, read a few back".

Read your block, again — 10 minutes

Two harder questions from your block page, with the tutor:

- *Which pins carry the **fastest** signal in your block?*
- *Where does your block's **ground** join the rest — and why only there?*

Then three minutes with the neighbour who will **review your PR**.

Reviewer ring: **A → B → C → D → E → A**.

KiCad — the only six operations

	Schematic	PCB
1	Place <code>A</code> — project library <code>class_board</code>	
2	Wire <code>w</code> , value <code>v</code> , fields <code>E</code> (footprint!)	
3	Annotate, ERC	
4	<i>Update PCB from Schematic</i> <code>F8</code> →	footprints land near your zone
5		Drag <code>M</code> , decoupling next to the pin
6		Route <code>x</code> (via <code>v</code>), DRC , 3D <code>Alt+3</code>

Layers: F.Cu signals • **In1.Cu = GND — never route on it** • In2.Cu power (instructor's)
• B.Cu signals.

Live on the projector now; V4 + V5 replay it.

E6 — schematic gaps (20 min)

1. `git switch -c b<N>-<name>` • open `hardware/class-board.kicad_pro` • your sheet `student/b<N>_gapped.kicad_sch`
2. Full PDF page for your block beside it. Your block page lists the **3–4 deleted parts**.
3. Each: place → wire → value → **footprint field** → keep the PDF's **reference designator**
4. **ERC → 0 errors.**

B1	one input network (AI4) + the ADC's AVDD decoupling
B2	one polyfuse + TVS pair (jack path) + 78L05 output caps
B3	AO2 difference-amp resistors + AO2 49.9 Ω / clamp
B4	one relay channel + one opto input network
B5	74HCT125 fast-out parts + one TTL terminal + buffer decoupling

E7 — route your zone (start, 30 min)

- F8 → the new footprints appear → drag them **inside** ZONE_B<N>, capacitors next to the pin they serve
- Route x : F.Cu first, B.Cu for crossings, **nothing on In1.Cu**, widths from the net classes (don't change them)
- **DRC every 15 minutes.** The two classics: *wrong layer, unconnected net*. Read every message.
- Block-specific: B1 nothing digital under the inputs • B2 1.0 mm for +5V, modules $\geq$ 15 mm from analog • B3 feedback resistors tight to the op-amp • B4 the 2.5 mm band • B5 FAST nets short with GND adjacent

Done today: **about half the zone.**

PR + peer review, live

1. `git add hardware/ && git commit && git push -u origin b<N>-<name>` → `gh pr create`
— **3D screenshot in the description**
2. *Reviewers* → your ring reviewer • CI must be green
3. Reviewer: *Files changed*, check out the branch, open the PCB, walk **the checklist**, $\geq$ 2 comments, *Approve / Request changes*

The checklist (chapter C, the instructor's merge list): ERC 0 / DRC 0 • `LCSC` on every part • values match the PDF • decoupling next to pins • nothing on In1.Cu • nothing outside the zone • pin 1 marked • ground pour joined • B4 band $\geq$ 2.5 mm • silkscreen on every connector • CI green

Homework 2 — 3 hours, gated

	h	Due
E7 finish routing — DRC 0, 3D screenshot (V5 first)	1.25	
E8 one part from LCSC — <code>easyeda2kicad --full --lcsc_id C...</code>	0.25	
E9 PR with the screenshot, request your reviewer	0.25	Wed 23 Sep
E10 review your peer (≥ 2 comments) • fix yours	0.5 • 0.25	Fri 25 • Sun 27
slack	0.4	

Mon 28 Sep: the instructor paste-merges the five zones, full DRC, Gerbers/BOM/CPL, quote, **order** — 5 assembled + 2 bare + 6 front panels. Boards back $\approx$ 12 Oct.

Fri 2 Oct: Firmware and Mechanical 101

Your block's driver and phone panel in sim mode • one alarm rule • your box from the template

No homework the week of 28 Sep. `/tutor HW2` until then.